Roy Models and Marginal Treatment effects

Romuald Méango

Department of Economics, University of Oxford

Table of Contents

Introduction

The Roy Model

Parametric Identification (Heckman, 1978)

MTE: Introduction, Definition and Justification

Relationship to other parameters

Estimation

A Checklist

Building bridges (Heckman, JEL 2010)

Policy evaluation problems:

P1 *"**Evaluating** the Impacts of Implemented Interventions on Outcomes Including Their Impacts on the Well-Being of the Treated and Society at Large"*

P2 *"**Forecasting** the Impacts (Constructing Counterfactual States) of Interventions Implemented in One Environment **in Other Environments**, including Impacts on Well-Being."*

P3 *"**Forecasting** the Impacts of Interventions (Constructing Counterfactual States Associated with Interventions) **Never Historically Experienced**, including Their Impacts on Well-Being."*

Table of Contents

Introduction

The Roy Model

Parametric Identification (Heckman, 1978)

MTE: Introduction, Definition and Justification

Relationship to other parameters

Estimation

A Checklist

Introduction to the Roy model

The Roy model puts an emphasis on the **economic intuition** behind the selection mechanism.

Sample selection models are closely linked to Roy Model (Historically, there was more interest in ‘missing data’ problem, what we now view as the ‘fundamental problem of causal inference’).

The selection mechanism gives an econometric formulation and economic intuition for the monotonicity assumption.

“Textbook” Model

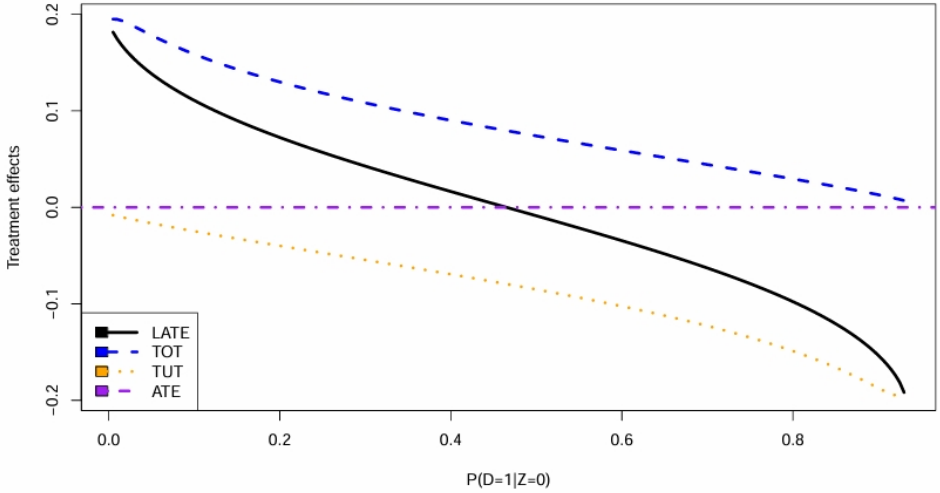
$$
\begin{aligned}
Y_0 &= \mu_0 + U_0 \\
Y_1 &= \mu_1 + U_1 \\
D &= 1\{\gamma_0 + \gamma_1 Z > V\} \\
Y &= (1 - D)Y_0 + DY_1
\end{aligned}
$$

$$
Z \sim \text{Bernoulli}(q) \quad \perp (V, U_0, U_1)
$$

$$
\begin{pmatrix} V \\ U_0 \\ U_1 \end{pmatrix} \sim \mathscr{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_0\rho_0 & \sigma_1\rho_1 \\ \sigma_0\rho_0 & \sigma_0^2 & \sigma_{01} \\ \sigma_1\rho_1 & \sigma_{01} & \sigma_1^2 \end{pmatrix} \right)
$$

Moving the window

First-stage effect 0.07, $q = 1/2$, $\delta = -0.1$



Relaxing Normality: the Latent Index Selection Model (LISM)

$$
\begin{aligned}
Y_0 &= \mu_0 + U_0 \\
Y_1 &= \mu_1 + U_1 \\
D &= \mathbf{1}\{\gamma_0 + \gamma_1 Z > V\} \\
Y &= (1 - D)Y_0 + DY_1, \\
Z &\sim \text{Bernoulli}(q) \perp\!\!\!\perp (V, U_0, U_1) \\
E[V] &= E[U_0] = E[U_1] = 0
\end{aligned}
$$

The Good:

- Simple model with instrument $Z \in \{0, 1\}$ and selection into treatment $D \in \{0, 1\}$.
- Treatment effects are heterogeneous and vary with "resistance" V.
- Do not assume joint normality of (U_0, U_1, V); mean zero WLOG.

Questions:

1. How does this compare to the LATE model?
2. Is this model identified? If so, can we estimate it?
3. If we can estimate it, does it allow us to go beyond LATE to ATE, TUT, TOT, etc.?

Assumptions of the Latent Index Selection Model

Treatment take-up

$$D(Z) = \mathbf{1}\{\gamma_0 + \gamma_1 Z > V\}.$$

Instrument relevance

$$P(\gamma_0 > V) \neq P(\gamma_0 + \gamma_1 > V).$$

Instrument exogeneity

$$Z \perp\!\!\!\perp (V, Y_0, Y_1).$$

- $\gamma_0 + \gamma_1 Z$ is the *latent index*.
- Relevance is what justified the Heckman two-step in the normal case.
- Exogeneity here: $Z \perp\!\!\!\perp (V, U_0, U_1)$ and $Y_j = \mu_j + U_j$ imply $Z \perp\!\!\!\perp (Y_0, Y_1, V)$.

(Slightly) Stronger Version of LATE Assumptions

Observable existence of compliers:

$$P(T = c) > 0 \quad \Leftrightarrow \quad E[D \mid Z = 1] \neq E[D \mid Z = 0].$$

No defiers in terms of potential treatments:

$$\text{Either } D_0 \leq D_1 \text{ or } D_1 \leq D_0 \text{ w.p.1.}$$

Replacement for mean exclusion (stronger):

$$Z \perp\!\!\!\perp (Y_0, Y_1, D_0, D_1)$$

which implies unconfounded type.

These two models are equivalent!

Latent Index Selection Model (LISM)

1. $D = \mathbf{1}\{\gamma_0 + \gamma_1 Z > V\}$.
2. $P(\gamma_0 > V) \neq P(\gamma_0 + \gamma_1 > V)$.
3. $Z \perp\!\!\!\perp (Y_0, Y_1, V)$.

Local Average Treatment Effects Model (LATE)

1. Either $D_0 \leq D_1$ or $D_1 \leq D_0$ w.p.1.
2. $E[D \mid Z = 1] \neq E[D \mid Z = 0]$.
3. $Z \perp\!\!\!\perp (Y_0, Y_1, D_0, D_1)$.

Comments:

- LISM assumptions imply LATE assumptions (straightforward).
- LATE assumptions imply LISM under additional technical conditions (see Glickman & Normand (2000), Vytacil (2002)).

Roy model vs. LATE Model (General case)

The LATE assumes a potential outcome framework for the D. For example, when Z is binary, with our notations:
$D(Z) = D(1)Z + D(0)(1 - Z)$, or more generally,

$$D(Z) = \sum_{z \in \mathscr{Z}} D(z) I\{Z = z\}$$

We require:

$$Z \perp\!\!\!\perp (\{Y(d) : d \in \mathscr{D}\}, \{D(z) : d \in \mathscr{D}\}) \text{ and } D(z') \geq D(z) \text{ if } z' \geq z.$$

Roy model vs. LATE Model

The Roy model explicitly models the choice problem. In the binary case:

$$\begin{aligned} D(Z) &= I\Big\{\mu_S(X,Z) - U_S \leq 0\Big\} \\ Y &= Y(1)D(Z) + Y(0)(1 - D(Z)) \end{aligned}$$

where:

- $\mu_S(X,Z) - U$ represents the 'utility' or 'return' of choosing $d = 1$ over $d = 0$;
- U_S is called a latent variable;
- $\mu_S(X,Z)$ is an unknown function (usually the average return at given X and Z).

Vytlacyl (2002) showed that this representation is equivalent to assuming monotonicity.

Roy model vs. LATE Model

The prototypical example is the study of returns to (college) education. In this case:

- D is the decision to attend college;
- Y is often the lifetime income (but could be over outcomes of interest);
- $\mu_S(X, Z) - U_S$ represents the perceived 'return' to college education and accounts both for pecuniary benefits ($Y_1 - Y_0$) and non pecuniary befits and cost (e.g. dis-utility of studying, tuition fees, consumption value of education, etc.).

Note that economic theory dictates the interpretation of the decision mechanism:

- $D = I(Y_1 > Y_0)$ (pure income maximization framework - Roy, 1951)
- or $D = I(Y_1 - Y_0 - C > 0)$, with perfect knowledge and C is a non-pecuniary cost .
- or $D = I(\mathbb{E}(Y_1 - Y_0 - C|\mathscr{I}) > 0)$, where $\mathscr{I}$ is the information set of the individual.

Sample Selection models

Sample selection models are often closely linked to the Roy model, because the 'missing data' is related to a participation decision.

For example:

- Labour market outcomes of workers: wages for non-workers are not observed when not working because of a self-selection in unemployment.

Notations:

Y_1^* Outcome of interest, and is observed if some $Y_s^* > 0$. For example:

- Y_s^* determines whether or not to work,
- Y_1^* determines how much to work,
- $Y_s^* \neq Y_1^*$: because of fixed costs to work such as commuting.

Sample Selection models

Bivariate sample selection model:

Participation equation:

$$D = \begin{cases} 1, & \text{if } Y_S^* = \mu_S(X,Z) - U_S > 0 \\ 0, & \text{if } Y_S^* = \mu_S(X,Z) - U_S \leq 0 \end{cases} \tag{1}$$

Outcome equation:

$$y_1 = \begin{cases} Y_1^*, & \text{if } Y_1^* > 0 \\ -, & \text{if } Y_1^* \leq 0 \end{cases} \tag{2}$$

Table of Contents

Introduction

The Roy Model

Parametric Identification (Heckman, 1978)

MTE: Introduction, Definition and Justification

Relationship to other parameters

Estimation

A Checklist

Roy models

The common specification of the Roy model is a linear model with normally distributed error. Define:

$$\begin{aligned} Y(0) &= X'\gamma_0 + U_0 \\ Y(1) &= X'\gamma_1 + U_1 \\ D &= I\{Y(1) - Y(0) + Z'\gamma + V\} \end{aligned}$$

with the parametric assumption that (U_0, U_1, V) are mean zero jointly normally distributed.

The ATE is identified by $ATE := \mathbb{E}(X'(\gamma_1 - \gamma_0'))$. Unfortunately, OLS is biased since D is correlated with the error term $U = U_1 D + U_0(1 - D)$ and Y.

Roy models

The observed outcome:

$$Y = X'\gamma_0 + DX'(\gamma_1 - \gamma_0) + U_0 + D(U_1 - U_0)$$

Note also that $D(U_1 - U_0)$ generates the heterogeneity in the effect of the treatment, even after controlling for observed characteristics. This heterogeneity is correlated with the selection mechanism:

$$D = I\left\{X'(\gamma_1 - \gamma_0) + Z'\gamma + V + (U_1 - U_0)\right\}$$

Note that $U_S := V + (U_1 - U_0)$ is normally distributed but is **not** independent of (U_1, U_0).

This model can be solved by Maximum Likelihood estimation.

Heckman's Two-Step Procedure

Consider $Y(1)$:

$$\begin{aligned} E(Y|X,Z,D=1) &= E(Y(1)|X,Z,X'(\gamma_1-\gamma_0)+Z'\gamma+U_S>0) \\ &= \mathbb{E}(X'\gamma_1+U_1|X,Z,X'(\gamma_1-\gamma_0)+Z'\gamma+U_S>0) \\ &= X'\gamma_1+E(U_1|X,Z,U_S>-(X'(\gamma_1-\gamma_0)+Z'\gamma)) \end{aligned}$$

Heckman (1979) notes that if $(\varepsilon_1,\varepsilon_2)$ are jointly normal:

$$\varepsilon_2=\sigma_{12}\varepsilon_1+\xi$$

where ξ is a random variable independent of ε_1.

Heckman's Two-Step Procedure

After some algebra, it follows that:

$$\begin{aligned} E(Y|X,Z,D=1) &= X'\gamma_1 - cov(U_1,U_S)\lambda(W'\gamma_s) \\ E(Y|X,Z,D=0) &= X'\gamma_0 + cov(U_0,U_S)\lambda(-W'\gamma_s) \end{aligned}$$

where $\lambda(z) = \phi(z)/\Phi(z)$: **inverse Mills ratio** (ϕ normal density function, Φ normal cumulative distribution function.)

Heckman's insight is that $\lambda(W'\gamma_s)$ enters as an additional regressor that 'controls' for the selection. Versions of this are known as the **control function** approach.

$cov(U_1,U_S)$ is an unknown parameter to be estimated.

Heckman's Two-Step Procedure

Heckit estimator:

1. Probit regression of D on (X, Z), to obtain $\hat{\gamma}_s$.
2. OLS model with the positive values of Y when $D = 1$ or $D = 0$:

$$Y_i = X'_{di}\beta_2 + \sigma_{ds}\lambda(W'_i\hat{\gamma}_s) + v_{di} \text{ if } D = d$$

Theoretically, the bivariate sample selection model with normal error is identified, even if there is no excluded variable Z.

However, the inverse Mills ratio is close to linear in some regions. Thus, because of concerns of multicollinearity, it is always advisable to have an excluded variable.

Heckman's Two-Step Procedure

Note that:

- Variance is heteroskedastic;
- Second OLS produces wrong standard errors (the valid standard error accounts for prediction $\hat{\gamma}_s$).
- $(H_0 : \sigma_{ds} = 0)$ is a test for the necessity of the correction.

The parametric version of the Roy model is **rarely used** in top-flight research today. There exists a nonparametric alternative.

Key assumption: Regression model for the error: $U_1 = \delta\varepsilon_s + \xi$ where $\xi \perp\!\!\!\perp \varepsilon_1$.

$$Y_i = X_{di}'\beta_2 + \sigma_{ds} h(W_i'\hat{\gamma}_s) + v_{di} \text{ if } D = d$$

and $h(z)$ is for example a non-parametric polynomial function.

Policy Relevant Treatment Effect

Question: What have we gained from modeling the choice decision?

Policy Relevant Treatment Effect

Beyond the identification of the ATE, the interest of the Roy model is to address additional counterfactual analysis as in problem **P2** and **P3**, as classified by Heckman (2010):

Define two policies.

- Policy "a": (Y_0^a, Y_1^a, C^a) and D_a
- Policy "b": (Y_0^b, Y_1^b, C^b) and D_b

Heckman and Vytlacyl (2001) define the Policy Relevant Treatment Effect:
$PRTE = \dfrac{\mathbb{E}(Y^b) - \mathbb{E}(Y^a)}{\mathbb{E}(D^b) - \mathbb{E}(D^a)}$ with $Y^a = D^a Y_1^a + (1 - D^a) Y_0^a$ and $Y^b = D^b Y_1^b + (1 - D^b) Y_0^b$.

$PRTE = ATE$ iff C^b is very small (universal participation) and C^a infinitely large (universal non participation).

Policy Relevant Treatment Effect

Assume there is policy invariance assumption: outcome is not changed by the policy, but selection is. For example, a is the current regime and b corresponds to a decrease of tuition by say 20 percent.

- $Y_0^a = Y_0^b, Y_1^a = Y_1^b$, but $C^a \neq C^b$:
- and there exists Z^a and Z^b, s.t.: $C^a = \mu_S(X, Z^a) + U_S$ and $C^b = \mu_S(X, Z^b) + U_S$.
- Thus: $Y^a = D^a Y_1 + (1 - D^a) Y_0$ and $Y^b = D^b Y_1 + (1 - D^b) Y_0$.

The Roy model estimation gives an estimate of

$$PRTE = \frac{\mathbb{E}_X(X')(\gamma_1 - \gamma_0)}{\mathbb{E}_X\big(P(D = 1|X, Z^b) - P(D = 1|X, Z^a)\big)}$$

Policy Relevant Treatment Effect

Insights:

- The PRTE is a LATE for an instrument that changes from Z^a to Z^b.
- If we have infinitemisal variation is Z, this would identifies some LATE for infinitesimal small variation of the instrument.
- The PRTE solves **P3:** "**Forecasting** the Impacts of Interventions **Never Historically Experienced**," as long as the selection mechanism remains stable, and we can extrapolate to some Z^b never historically experienced.
- The interpretation of the selection mechanism is crucial in this analysis.

Heckman's Two-Step Procedure

Sample Selection package in R: `https://cran.r-project.org/web/packages/sampleSelection/sampleSelection.pdf`

An example: `https://rpubs.com/hacamvan/316839`

Table of Contents

Introduction

The Roy Model

Parametric Identification (Heckman, 1978)

MTE: Introduction, Definition and Justification

Relationship to other parameters

Estimation

A Checklist

MTE

As we will see, the MTE generalises the approach and insights from the Roy Model.

It shows that there exists a control function that can be introduced to control for endogeneity.

This control function depends on the propensity score, i.e. the probability to be treated.

In the parametric case, one could write a close form solution of the control function (inverse Mills ratio). Without the parametric assumption, there is no closed form solution and we rely on nonparametric/semiparametric methods.

The MTE is **more than a nonparametric Roy**, it provides a unifying framework for policy evaluation.

MTE and Evaluation Parameters

Different evaluation parameters are an average over parts of the distribution of impacts.

- The ATE averages over the **entire** distribution.
- The ATT averages over the distribution of impacts for those who are somehow **allocated into treatment**.
- LATE averages over the distribution of impacts for those who **switch into treatment** as a result of a reform or more precisely, as a result of a change of the value of some instrument affecting decisions to participate.

Thus they all represent an aggregation over different margins. As such they are **not comparable** and sometimes difficult to interpret.

MTE and Evaluation Parameters

As a unifying parameter, Heckman and Vytlacil (2005) defined the Marginal Treatment Effect.

This is the effect of a treatment on the marginal individual entering treatment. The MTE will provide an interpretation of several evaluation parameters.

They will provide a '**bridge**' between structural and treatment effect parameters and allow us to understand the way they are related.

MTE: A definition

Consider a discrete treatment D. The true allocating to treatment may be written as:

$$D_i = I(U_{si} \leq Z_i' \gamma_s)$$

For a particular value of $Z_i' \gamma_s$, the marginal individual is the one with $U_{si} = Z_i' \gamma_s$.

Now consider the effect of treatment for the individual i:

$$\tau_i = Y_i(1) - Y_i(0).$$

The Marginal Treatment Effect is defined by

$$MTE(Z_i' \gamma) = \mathbb{E}\left(\tau_i | U_{si} = Z_i' \gamma_s\right).$$

In the textbool model, varying $\gamma_1 \to 0$, gives $V = \gamma_0$.

MTE: A definition

The Marginal Treatment Effect is defined by

$$MTE(Z_i'\gamma) = \mathbb{E}\left(\tau_i | U_{si} = Z_i'\gamma_s\right).$$

Thus the *MTE* is the average impact for the marginal individual receiving treatment among those with value of the index equal to $Z_i'\gamma_s$.

It turns out that all parameters (ATE, LATE, ATT, ATNT) can be written as weighted averages of this parameter.

MTE: A simple justification

Consider a very simple model of College attendance choice.

The Cost of attending is $C_i = Z_i'\gamma + \varepsilon_{si}$ where C_i encompasses the opportunity cost in lost earnings and the direct costs.

The benefits in the simplest form are $Y_i(1) - Y_i(0)$.

The decision to attend college is given by:

$$D_i = 1 \Leftrightarrow Y_i(1) - Y_i(0) \geq Z_i'\gamma + \varepsilon_{si}$$

MTE: A simple justification

The marginal individual satisfies the condition:

$$
\begin{aligned}
Y_i(1) - Y_i(0) - Z_i'\gamma = \varepsilon_{si} \quad &\Leftrightarrow \quad \mu_1(Z_i) - \mu_0(Z_i) - Z_i'\gamma = U_0 - U_1 + \varepsilon_{si} \\
&\Leftrightarrow \quad \mu_1(Z_i) - \mu_0(Z_i) - Z_i'\gamma := V_{si}
\end{aligned}
$$

This shows how the allocation to treatment will depend on the returns and why conditioning on V_s will give us the treatment effect for the marginal individual at a given Z_i.

MTE: A simple justification

Consider the case where outcomes are linear, and $\mu_1(Z_i) = X_i'\gamma_0$, with X a sub-vector of Z.

$$\begin{aligned} Y(0) &= X'\gamma_0 + U_0 \\ Y(1) &= X'\gamma_1 + U_1 \end{aligned}$$

And consider the observed outcome:

$$Y = X'\gamma_0 + DX'(\gamma_1 - \gamma_0) + U_0 + D(U_1 - U_0)$$

It is convenient to rewrite the treatment model as:

$$D_i = I\left(U_{si} \leq F_V(Z_i'\gamma_s)\right)$$

where U_{si} is uniform $U[0,1]$, where F_V is the distribution function of V_s and $P(Z_i) = F(Z_i'\gamma_s)$.

MTE: A simple justification

The key step is to consider the probability of assignment to treatment P(Z) as an instrument.

Assumption (Independence)

Assume that:

$$(U_0, U_1, U_s) \perp\!\!\!\perp P(Z) | X$$

Interpretation: The treatment assignment is independent of the private gains.

MTE: A simple justification

Assumption (Support)

Given X, $P(Z)$ has continuous support in the open interval $(0,1)$.

Interpretation: the excluded variables Z have at least one continuous component and Z varies sufficiently for any fixed value of X to make the treatment assignment probability vary anywhere between 0 and 1.

MTE: A simple justification

The expected value of the outcome given the instrument $P(Z) = p$ and X:

$$\mathbb{E}(Y|X, P(Z) = p) = X'\gamma_0 + P(Z)X'(\gamma_1 - \gamma_0) + U_0 + E\big(D(U_1 - U_0)|X, P(Z) = p\big)$$

Given the allocation mechanism, $D = 1$ over the interval $u_s \in [0, p]$ and $D = 0$ for $u_s \in (p, 1]$. Hence:

$$E\big(D(U_1 - U_0)|X, P(Z) = p\big) = \int_{-\infty}^{+\infty} \int_0^p (U_1 - U_0) f(U_1 - U_0|X, U_s = u_s)\, du_s d(U_1 - U_0)$$

MTE: A simple justification

Take the derivative with respect to p:

$$\begin{aligned}
\frac{\partial E(Y|X, P(Z)=p)}{\partial p} &= X'(\gamma_1 - \gamma_0) + \int_{-\infty}^{+\infty} (U_1 - U_0) f(U_1 - U_0|X, U_s = p)\, d(U_1 - U_0) \\
&= X'(\gamma_1 - \gamma_0) + E(U_1 - U_0|X, U_s = p) \\
&= E(Y_1 - Y_0|X, U_s = p)
\end{aligned}$$

$E(U_1 - U_0|X, U_s = p)$ is the average unobserved gain of treatment for those whose unobserved characteristics make them indifferent between treatment or not at $P(Z) = p$.

MTE: A simple justification

$\dfrac{\partial E(Y|X, P(Z) = p)}{\partial p}$ is known as the 'local instrumental variables'. It is the **limit** of the Wald estimator, for a marginal variation of $P(Z)$.

$E(Y_1 - Y_0|X, U_s = p)$ is the marginal treatment effect for individuals with observable characteristic X and at the margin of participation $U_s = p$.

Table of Contents

Introduction

The Roy Model

Parametric Identification (Heckman, 1978)

MTE: Introduction, Definition and Justification

Relationship to other parameters

Estimation

A Checklist

MTE in relationship to ATE

We can now define all parameters of interest as a function of the MTE.

Averaging over the support of marginal individuals:

$$
\begin{aligned}
ATE(X) &= \int_0^1 MTE(p)dp && (3)\\
&= X'(\gamma_1 - \gamma_0) + \int_0^1 E(U_1 - U_0 | u_s)du_s && (4)\\
&= X'(\gamma_1 - \gamma_0) && (5)
\end{aligned}
$$

MTE in relationship to LATE

The LATE is ATE for those induced to change treatment assignment because of Z.
More formally, for any given X, P(Z) takes two values, say $P(z^*) = b(X)$ or $P(z^{**}) = a(X)$ with say $a > b$.

$a(X) - b(X)$ the change in the probability people to take treatment for an individual with characteristics X, as induced by the change in the instrument.

All those with unobserved propensity to be assigned to treatment such that $b(X) \leq u_s \leq a(X)$ will now switch into treatment.

MTE in relationship to LATE

Averaging over the support of marginal individuals:

$$LATE(X) = \frac{1}{a(X)-b(X)} \int_{b(X)}^{a(X)} MTE(p)dp \tag{6}$$

$$= \frac{1}{a(X)-b(X)} \left(X'(\gamma_1 - \gamma_0) + \int_{b(X)}^{a(X)} E(U_1 - U_0 | u_s) du_s \right) \tag{7}$$

LATE is the average MTE with weights $\frac{1}{a(X)-b(X)}$ in the range $(a(X), b(X))$ and zero everywhere else.

Note that the value of LATE will depend on the interval over which we integrate, i.e. it will depend on which margin the specific policy tend to shift into treatment.

MTE in relationship to ATT

Consider now the parameter Average Treatment on the Treated. This can be written as:

$$ATT(X) = \int_0^1 MTE(p) \left[\frac{\Pr\left(P(Z|X) > p\right)}{E\left(P(Z|X)\right)} \right] dp$$

Note that the higher the probability of participating, the larger the weight on the overall effect.

Table of Contents

Introduction

The Roy Model

Parametric Identification (Heckman, 1978)

MTE: Introduction, Definition and Justification

Relationship to other parameters

Estimation

A Checklist

MTE: Estimation

The above process suggests an estimation procedure:

1. Estimate the nonparametric regression of D on Z, $P(Z)$;
2. Estimate the nonparametric regression of the outcome variable Y on X and on $P(Z)$, e.g., fitting Y on polynomials of X and $P(Z)$;
3. Differentiate the result with respect to P(Z)

If $P(Z)$ indeed varies from $(0, 1)$, i.e. has full support, then it will be possible to estimate the marginal treatment effect on the whole support.

Theory versus Practice

- Local IV involves nonparametric regression of Y on X and the propensity score.
- This is extremely challenging in practice even if X is low-dimensional.
- Need variation in the propensity score for fixed X; this comes from the instrument Z.
- To nonparametrically identify the full MTE function, need an instrument that allows $P(Z)$ to vary over the full range $[0, 1]$ for any value of X.

Semiparametric Estimation of MTE

Estimation of MTE is usually done semiparametrically. The key is a parametrisation of the 'moment response function'. For example, the linear model.

$$\begin{aligned} m_d(u,x) &= E(Y_d|U=u,X=x) = \alpha_d + \beta_d x + \gamma_d u \\ &\Rightarrow E(Y|D=d,X=x,U=u) = \alpha_d + \beta_d x + \frac{\gamma_d}{2} u \end{aligned}$$

or the saturated model:

$$\begin{aligned} m_d(u,x) &= E(Y_d|U=u,X=x) = \alpha_d + \beta_d x + \gamma_d u + \delta xu \\ &\Rightarrow E(Y|D=d,X=x,U=u) = \alpha_d + \beta_d x + \frac{\gamma_d}{2} u + \frac{\delta_d}{2} xu \end{aligned}$$

Semiparametric Estimation of MTE

The above process suggests an estimation procedure:

1. Estimate the nonparametric regression of D on Z, to obtain $P(Z)$;
2. Regress Y on x, p in the linear model (on x, p, xp in the saturated model) and collect the coefficients;
3. The estimator of the $MTE(x, p)$ is

$$\hat{E}(Y_1 - Y_0|x, p) = \hat{\alpha}_1 - \hat{\alpha}_0 + (\hat{\beta}_1 - \hat{\beta}_0)x + (\hat{\gamma}_1 - \hat{\gamma}_0)p \text{ (linear model)}$$
$$\hat{E}(Y_1 - Y_0|x, p) = \hat{\alpha}_1 - \hat{\alpha}_0 + (\hat{\beta}_1 - \hat{\beta}_0)x + (\hat{\gamma}_0 - \hat{\gamma}_0)p + (\hat{\delta}_1 - \hat{\delta}_0)xp \text{ (Saturated model)}$$

4. Estimate your target parameter, e.g. $\widehat{ATE} = \hat{\alpha}_1 - \hat{\alpha}_0 + (\hat{\beta}_1 - \hat{\beta}_0)x$

Estimation of MTE

Parametrisation reflects a trade-off between flexibility of the model and conditions on the instrument.

The linear model requires two points of support of the instrument.
The saturated model requires two points of support of the instrument **at each value of** x.

The command 'ivmte' in the R-package 'ivmte' implements the linear model.

To go further on this topic: Brinch et al. (2017); Mogstad et al. (2018)

Cornelissen et al. (QJE, 2018) — Background

Background

- Major policy question: causal effect of early childhood interventions, including state-provided day-care.
- Some targeted-program studies (e.g. Head Start) find sizable positive effects; evidence for universal provision is mixed (e.g. Quebec).
- Possible explanation: targeting enrolls those most likely to benefit — how to reconcile mixed evidence?
- Hypothesis: targeted programs enroll children with worse home environments who would gain most.

Cornelissen et al. — This study

- Study universal preschool/childcare provision in Germany using the MTE approach.
- Treatment: early attendance (at least three years).
- Instrument: staggered roll-out of 1990s policy reform that changed slot availability across locations.
- Outcome: universal school-readiness exam at age 6.

Cornelissen et al. — Main findings

- Evidence of reverse selection on gains from observed characteristics.
- Minorities benefit most from childcare but are least likely to enroll.
- Similar selection on unobservables: “high resistance” children benefit most.
- Effect so strong that ATNT $>$ ATE $>$ 0 $>$ ATT.
- Evidence that treatment-effect heterogeneity comes from Y_0 rather than Y_1.

A simplified MTE model

Additive separability

$$E[U_0 \mid U_S, X] = E[U_0 \mid U_S], \qquad E[U_1 \mid U_S, X] = E[U_1 \mid U_S]$$

- Changing X only affects the intercept of the MTE (as a function of U_S).
- Still allows U_S to vary with X.

Implications of separability and linearity

MTE function

$$\begin{aligned}
\mathrm{MTE}(p,x) &= \mu_1(x) - \mu_0(x) + E(U_1 - U_0 \mid X = x, U_S = p) \\
&= \mu_1(x) - \mu_0(x) + E(U_1 - U_0 \mid U_S = p) \quad \text{(Separability)} \\
&= x^\top(\beta_1 - \beta_0) + E(U_1 - U_0 \mid U_S = p) \quad \text{(Linearity)} \\
&= x^\top(\beta_1 - \beta_0) + \frac{d}{dp}K(p)
\end{aligned}$$

Observed conditional mean:

$$E[Y \mid P(X,Z) = p, X = x] = x^\top\beta_0 + x^\top(\beta_1 - \beta_0)p + K(p).$$

This is a semi-parametric model: linear regression plus unknown $K(p)$.

A parametric approximation

- Instead of fully nonparametric $K(p)$, model $K(p)$ as a polynomial in p (exclude constant and linear terms already captured by the regression).
- Specification used (example):

$$E[Y \mid P(X,Z) = p, X = x] = x^\top \beta_0 + x^\top (\beta_1 - \beta_0) p + \sum_{j=2}^{J} \alpha_j p^j.$$

- If we knew true p we could run this regression; but we don't observe $p = P(X,Z)$ directly.

Implementation (practical steps)

1. Run a probit/logit of D_i on (X_i, Z_i) to estimate propensity scores $\widehat{p}_i$.
2. Estimate β_0, β_1, α from the regression

$$Y_i = X_i\beta_0 + X_i^\top(\beta_1 - \beta_0)\widehat{p}_i + \sum_{j=2}^{J} \alpha_j \widehat{p}_i^j + \varepsilon_i.$$

3. Construct the estimated MTE:

$$\widehat{\mathrm{MTE}}(p, x) = \frac{\partial}{\partial p}\left[x^\top \widehat{\beta}_0 + x^\top(\widehat{\beta}_1 - \widehat{\beta}_0)p + \sum_{j=2}^{J} \widehat{\alpha}_j p^j\right].$$

4. Take weighted averages of $\widehat{\mathrm{MTE}}(p, x)$ to form ATE, PRTEs, LATEs, etc.

Some specifics (Cornelissen et al., 2018)

- Add municipality (R) and exam-cohort (T) dummies:

$$Y = X\beta_0 + \alpha_R + \tau_T + X(\beta_1 - \beta_0)\, p_b + \sum_{j=1}^{J} \alpha_j p_b^j + \varepsilon.$$

- Experiment with $J = 2, 3, 4$ and with a semi-parametric specification.
- Reminder: differentiation yields the MTE, so $J = 2$ corresponds to a linear specification for $E(U_1 - U_0 \mid U_S)$.
- In their application, results were similar across different choices of $K(p)$.

MTE

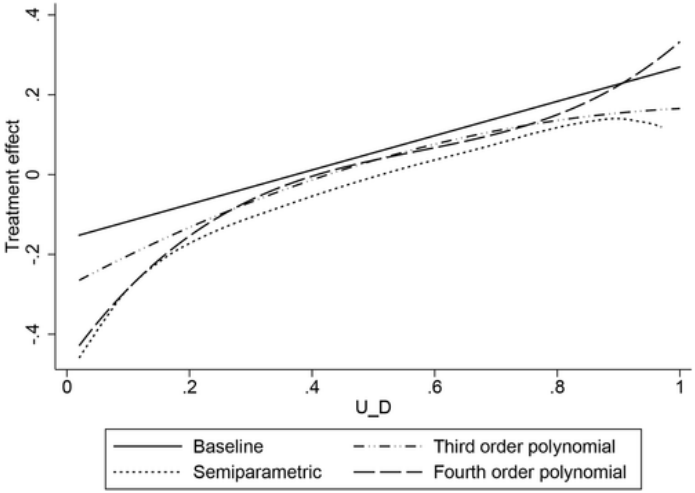


Figure: MTE as a function of P - Different specifications

PRTE

TABLE 9
POLICY-RELEVANT TREATMENT EFFECTS

	PRTE (1)	Propensity Score: Baseline (2)	Propensity Score: Policy (3)
1. Bring 2002 $P(Z)$ to .9 by adding .275	.160* (.085)	.67	.90
2. Bring 2002 $P(Z)$ to .9 by multiplying 1.5	.165* (.087)	.67	.90
3. Lift 2002 cohort's coverage rate (Z) to 1 if < 1	.123 (.077)	.67	.71
4. Add .4 to 2002 cohort's coverage rate (Z)	.141* (.086)	.67	.72

Table of Contents

Introduction

The Roy Model

Parametric Identification (Heckman, 1978)

MTE: Introduction, Definition and Justification

Relationship to other parameters

Estimation

A Checklist

IV, LATE, MTE: A checklist I

This provides you with a series of questions to ask and steps to perform before taking modeling decisions. This starts as before:

1. Classify the variables in your dataset:
   - Which ones are predetermined and can serve as controls?
   - Which ones can be thought of as intermediate outcomes and serve as explanatory mechanisms or mediators?
   - Is there the possibility to use one for a placebo test, e.g. values of the outcome before treatment?
   - Is there an instrumental variable in my dataset? What are the identifying assumptions?How plausible are these assumptions?

IV, LATE, MTE: A checklist II

2. Check your data:
   - Always do a data summary.
   - How is the balance of covariates between treatment and untreated? how does it compare to the balance across the instrument?
   - Do I have a first stage? Does it look as if I have an ITT different from zero?
   - Is it plausible to assume monotonicity? If not, is it plausible to assume treatment effect homogeneity?
   - After controlling on covariates, is there still a first-stage?
3. Estimate your effect and characterise the population of compliers
   - What is relevant to my study (LATE, PRTE, MTE, ATE)?
   - How large is the complier population?
   - How representative is the complier population?
   - Is there TE heterogeneity w.r.t observable characteristics?

IV, LATE, MTE: A checklist III

What speaks for the LATE?

- Binary instrument and binary treatment,
- the application is one of the special cases where the LATE identifies the ATT or ATNT (e.g., one-sided non-compliance),
- The population of compliers seems representative,
- The goal of the study is somewhat modest, and can satisfy itself with a local effect (e.g. there is no or little prior evidence of this effect).

When implementing, remember:

- to present and discuss the characteristics of the compliers,
- If you have several instruments, compare the results,
- In the case with binary instruments, avoid TSLS as there are good feasible alternative.

IV, LATE, MTE: A checklist IV

What speaks for TSLS?

- I am after an homogeneous treatment effect, or can saturate the model,
- I have a rather complicated model with many covariates and possibly multiple instruments (e.g. IV panel data),
- The goal of the study is somewhat modest, and the literature has so far satisfied itself with an homogeneous treatment effect (e.g., all prior evidence are from OLS or TSLS).

IV, LATE, MTE: A checklist V

When implementing, remember:

- to check for outliers.
- to assess heterogeneity with respect to observable characteristics.
- Keane and Neal (2021) suggest that the rule-of-thumb of an F-stat > 10 is too weak a requirement.
- They also suggest to report the Anderson-Rubin (AR) test and confidence interval in lieu of the $t - stat$. See their paper: `https://www.sciencedirect.com/science/article/pii/S0304407623000222`

IV, LATE, MTE: A checklist VI

What speaks for the MTE framework?

- A continuous instrument with a decent first stage (propensity score ranging on a large part of the unit interval),
- I am fine with doing some extrapolation from my limited observation of the propensity score,
- It is important for the study to learn about individual selection, for example, selection on gains,
- The goal of the study is to infer a PRTE.

IV, LATE, MTE: A checklist VII

When implementing:

- A nice addition of MTE, not really feasible with LATE is to show the effect of unobservable of gains. Be sure to plot $m_1(u,x), m_0(u,x)$ and $m_1(u,x) - m_0(u,x)$. Discuss the heterogeneity (e.g., Heckman, Urzua, and Vytlacyl, 2006).
- Be parsimonious with the specification of the moment response function.

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